

Foundations of Cybersecurity / Applied Cryptography

21 February 2023

DOCENTE

1. Illustrate and discuss the minimum set of information items you expect to find in a certificate.
2. Explain the difference between an invalid and a revoked certificate.
3. For each of the following certificate templates, indicate whether it is secure or not. Motivate your answers.
 - a. $\langle A, \Pi_A, L_A \rangle_A$
 - b. $\langle A, \Pi_A, L_A \rangle_{CA}$
 - c. $\langle A, \Pi_A, L_A \rangle_B, \langle B, \Pi_B, L_B \rangle_B$
 - d. $\langle A, \Pi_A, L_A \rangle_B, \langle B, \Pi_B, L_B, ca = yes \rangle_{CA}$

Where $S_P(X)$ denotes the digital signature of principal P on statement X ; $\langle X \rangle_P$ denotes $\{X, S_P(X)\}$; Π_P denotes principal P 's public key, L_P denotes the validity interval of Π_P ; and finally, CA denotes a trusted certification authority.

Exercise n. 2 [12]

Assume that a file (possibly the whole volume) is stored in its encrypted form under AES-128-CBC. The file can be randomly accessed on a per-block basis.

1. Show that it is possible to apply a controlled modification to a given location of the plaintext file while operating on the encrypted version (malleability attack).
2. Argue the price the adversary must pay for this attack.
3. Assume that we insert a parity bit in each file block. Is this control able to mitigate or prevent the malleability attack?

Exercise n. 3 [6]

Find, explain and patch vulnerabilities, if any, of the following function:

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(a+b>UINT_MAX)
        return 0;
    return a+b;
}
```

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Solution

Exercise n.1

1. See theory.
2. See theory.
3. An answer for each template follows.
 - a. Insecure: self-signed certificate.
 - b. Secure: signed by a trusted CA.
 - c. Insecure: Bob hasn't been delegated.
 - d. Secure: Bob has been delegated (ca = yes) by a trusted CA.

Exercise n.2

Questions 1 and 2. According to CBC, $p_i = D_k(c_i) \oplus c_{i-1}$. Therefore, if you modify c_{i-1} into c'_{i-1} , you get:

- a) $p'_{i-1} = D_k(c'_{i-1}) \oplus c_{i-2}$, and
- b) $p'_i = D_k(c_i) \oplus c'_{i-1}$

While p'_{i-1} is completely lost (the price to pay), bits of p'_i commute where c'_i bits are commuted w.r.t. to c_i .

Question 3. A parity bit does not help because the adversary can commute it as needed according to the number of other bits of the block that are commuted.

Exercise n.3

The problem is wraparound. The error condition is poorly and consequently the *if* clause expression will never be true, not even in the case $a + b$ wraps around. The following patch must be applied.

```
unsigned int do_stuff(unsigned int a, unsigned int b){
    if(a > UINT_MAX-b)
        return 0;
    return a+b;
}
```